Glycolysis: Complete Molecular Reaction Map

Ten enzyme-catalysed reactions, molecular structures, phosphate transfers, and redox accounting

Pathway: Glucose → 2 Pyruvate

Net yield per glucose: 2 ATP, 2 NADH, 2 pyruvate

Structural notation

- Blue: phosphate transfer or phosphate-bearing group.
- Red: oxidation-reduction bookkeeping.
- Green: enzyme label.
- Carbon skeletons are shown in open-chain form.
- Protonation states are schematic; physiological forms depend on pH and the ionic environment.

Important: The displayed structural formulas are intended to show connectivity and functional-group changes. They omit stereochemistry and do not represent every enzyme-bound intermediate.

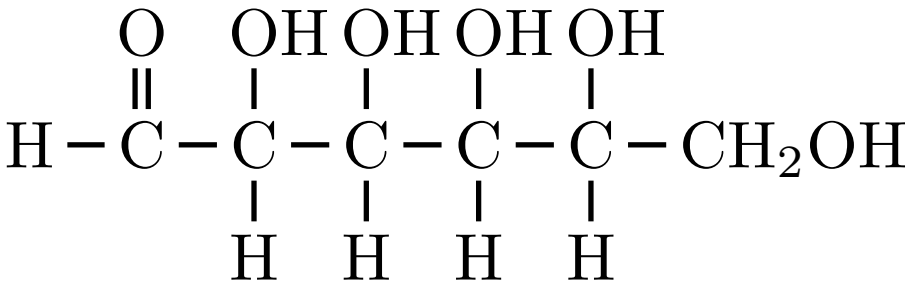
PHASE I: ENERGY INVESTMENT

STEP 1. Glucose → Glucose-6-phosphate

Enzyme: Hexokinase

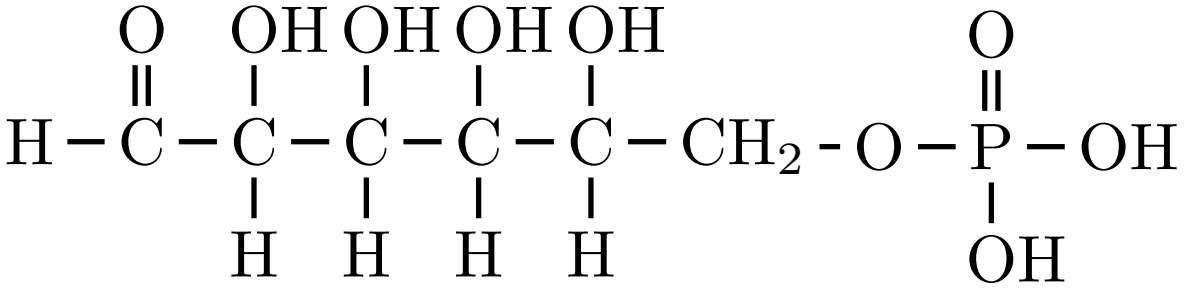
$$\text{Glucose} + \text{ATP} \longrightarrow \text{Glucose}{-}\text{6-phosphate} + \text{ADP}$$

Glucose:



$$\xrightarrow[Mg^{2+}]{\text{Hexokinase; ATP}\rightarrow\text{ADP}}$$

Glucose-6-phosphate:



Chemical changes

1. The oxygen of the C6 hydroxyl group participates in phosphoryl transfer.
2. The terminal phosphoryl group of ATP is transferred to the C6 oxygen.
3. The C6 group changes from $-CH_2OH$ to $-CH_2OPO_3H_2$ in the schematic formula.
4. ATP becomes ADP.

Transfer:

$$\text{ATP} \longrightarrow \text{ADP} + \text{phosphate transferred to glucose}$$

Redox: No net oxidation or reduction occurs.

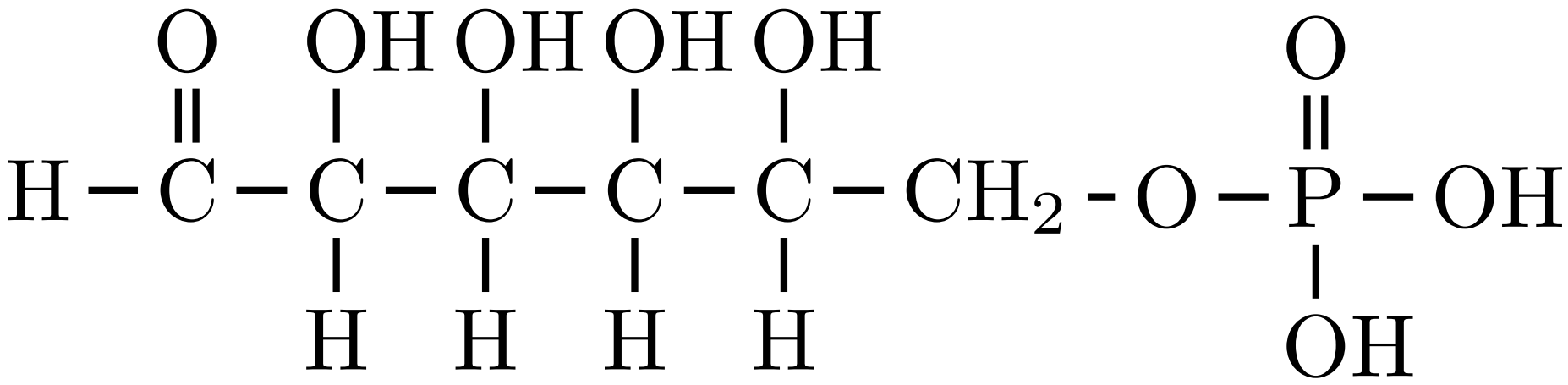
Atom accounting: All six glucose carbon atoms are retained.

ATP consumed: 1

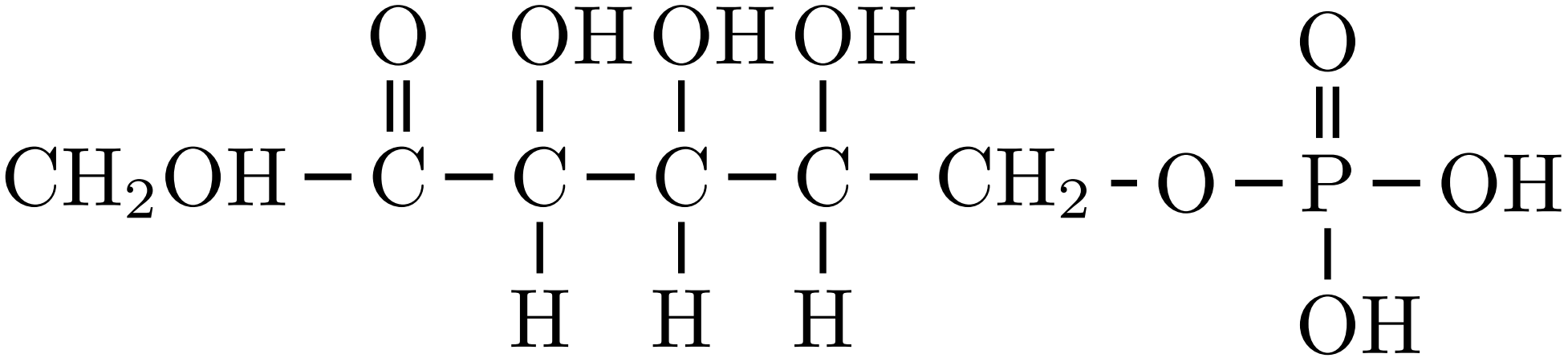
STEP 2. Glucose-6-phosphate → Fructose-6-phosphate

Enzyme: Phosphoglucose isomerase

$$\text{Glucose}-6\text{-phosphate} \rightleftharpoons \text{Fructose}-6\text{-phosphate}$$



Phosphoglucose isomerase



Chemical changes

1. The cyclic sugar can open into its linear form.
2. The aldehyde functionality at C1 is converted into a ketone functionality at C2.

3. The reaction proceeds through an enediol or enediolate-like intermediate.
4. Proton transfer and carbonyl reformation yield fructose-6-phosphate.

Electron movement: Bond electrons are redistributed within the sugar. There is no net electron transfer to NAD^+.
Phosphate accounting: The phosphate remains attached at C6.

$$\boxed{\text{ATP consumed: } 0}$$

STEP 3. Fructose-6-phosphate → Fructose-1,6-bisphosphate

Enzyme: Phosphofructokinase-1 (PFK-1)

$$\text{Fructose}-6\text{-phosphate} + \text{ATP} \longrightarrow \text{Fructose}-1,6-\text{bisphosphate} + \text{ADP}$$

$$\boxed{\text{F}_6\text{P} + \text{ATP} \xrightarrow[\text{Mg}^{2+}]{\text{PFK}-1} \text{F}_{1,6}\text{BP} + \text{ADP}}$$

Chemical changes

1. The C1 hydroxyl group is phosphorylated.
2. ATP donates a terminal phosphoryl group.
3. The pre-existing phosphate at C6 is retained.
4. The product now has phosphate groups at C1 and C6.

Transfer:

$$\text{ATP} \longrightarrow \text{ADP}$$

A phosphate group is added to the C1 oxygen of the sugar.
Redox: No net oxidation-reduction reaction occurs.

$$\boxed{\text{ATP consumed: } 1}$$

$$\boxed{\text{Cumulative ATP investment: } 2}$$

STEP 4. Fructose-1,6-bisphosphate cleavage

Enzyme: Aldolase

$$\text{Fructose}-1,6-\text{bisphosphate} \rightleftharpoons \text{GAP} + \text{DHAP}$$

Six-carbon substrate

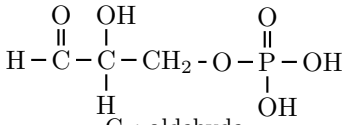
Fructose-1,6-bisphosphate

C_6; phosphates at C1 and C6

Aldolase

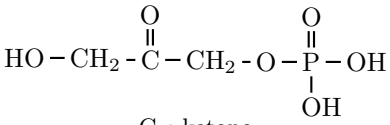
Cleaves the C3–C4 bond

Glyceraldehyde-3-phosphate



C_3; aldehyde

Dihydroxyacetone phosphate



C_3; ketone

Chemical changes

1. The six-carbon substrate undergoes carbon-carbon bond cleavage.
2. The bond between C3 and C4 is broken.
3. One three-carbon product is GAP.
4. The other three-carbon product is DHAP.
5. Both products contain a phosphate group.

Electron movement: Bond electrons are redistributed during the cleavage mechanism. The net reaction is not an oxidation-reduction reaction.

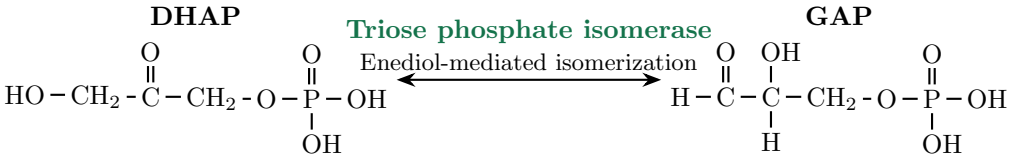
$$\boxed{C_6 \longrightarrow C_3 + C_3}$$

No carbon dioxide is released.

STEP 5. DHAP ⇌ GAP

Enzyme: Triose phosphate isomerase

Dihydroxyacetone phosphate ⇌ Glyceraldehyde−3-phosphate



Chemical changes

1. The ketone functionality of DHAP is converted into the aldehyde functionality of GAP.
2. A proton is transferred through the enzyme-catalysed mechanism.
3. The three-carbon skeleton and phosphate group are retained.

The GAP generated in step 4 remains in the pathway. The DHAP generated in step 4 is converted into another GAP.

Two GAP molecules per original glucose

No ATP consumed or produced

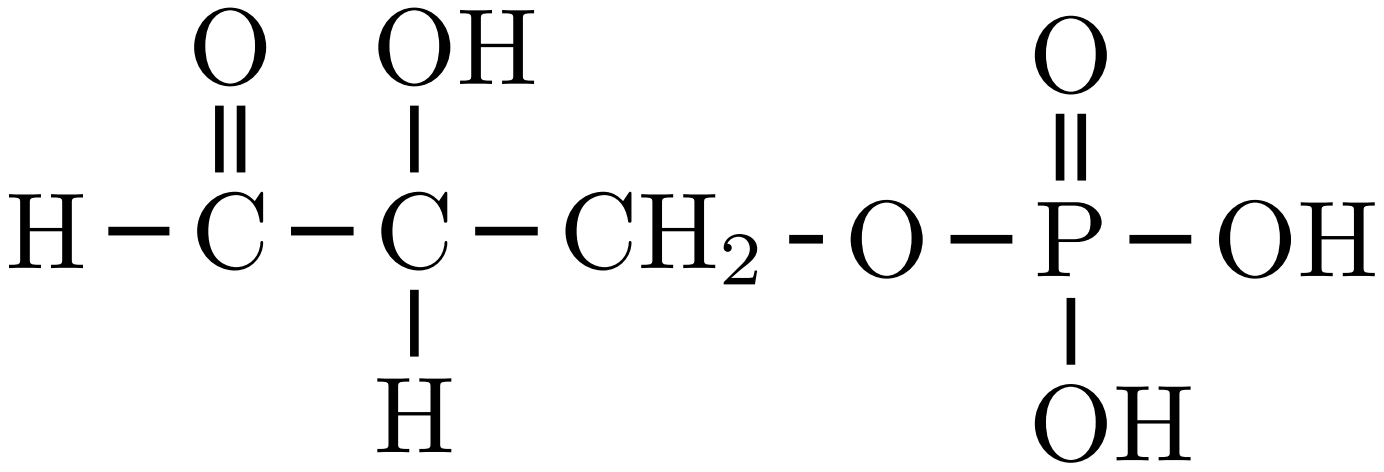
PHASE II: ENERGY PAYOFF

STEP 6. GAP oxidation and formation of 1,3-BPG

Enzyme: Glyceraldehyde-3-phosphate dehydrogenase (GAPDH)

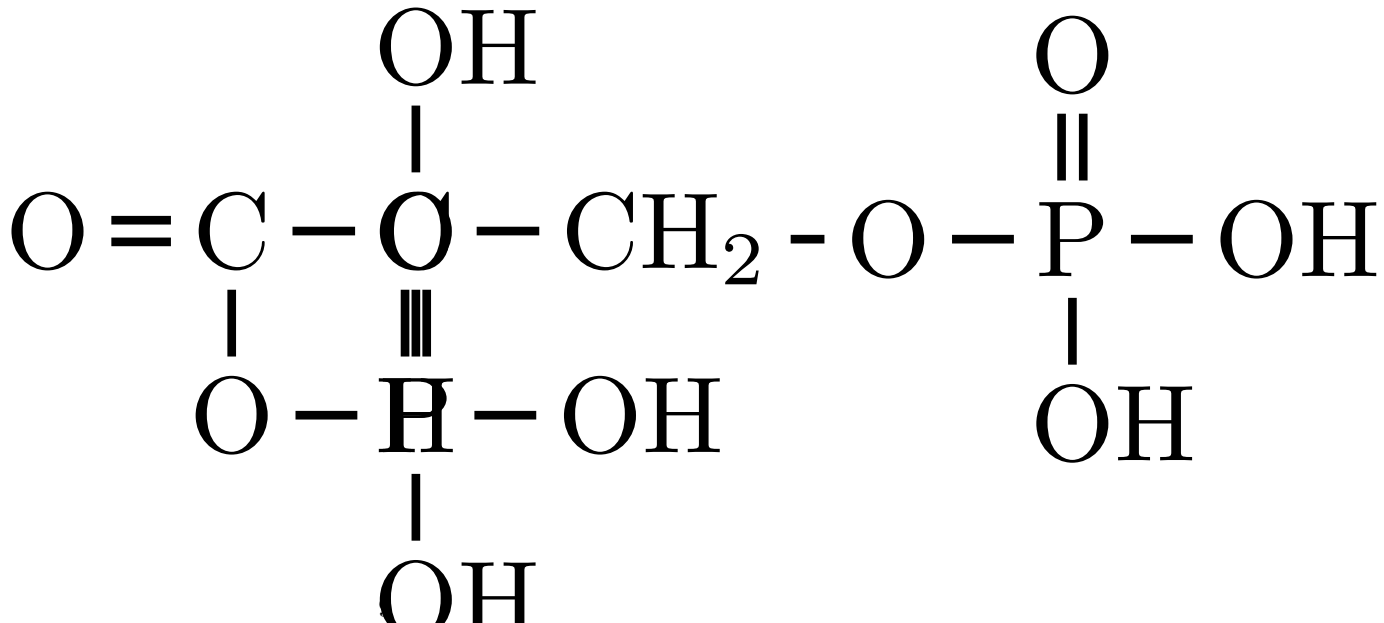
$$\mathrm{GAP} + \mathrm{NAD^+} + \mathrm{P}_i \rightleftharpoons \mathrm{1{,}3{-}BPG} + \mathrm{NADH} + \mathrm{H^+}$$

GAP:



$\xrightleftharpoons{\text{GAPDH; NAD}^+,\ P_i}$

1,3-BPG:



Chemical changes

1. The aldehyde-derived carbon at C1 is oxidized.
2. NAD^+ accepts reducing equivalents and becomes NADH.
3. Inorganic phosphate is incorporated into the C1 acyl-phosphate group.
4. The phosphate already attached at C3 remains in place.

Phosphate origin

$$\boxed{\underbrace{\text{C1 phosphate}}_{\text{from inorganic } P_i} \qquad \underbrace{\text{C3 phosphate}}_{\text{retained from GAP}}}$$

Electron transfer

$$\text{NAD}^+ + \text{reducing equivalents} \longrightarrow \text{NADH}$$

The oxidation of GAP supplies the reducing equivalents. NAD^+ is reduced to NADH.

$$\boxed{2\,\text{GAP} + 2\,\text{NAD}^+ + 2\,\text{P}_i \longrightarrow 2\,\text{1,3–BPG} + 2\,\text{NADH} + 2\,\text{H}^+}$$

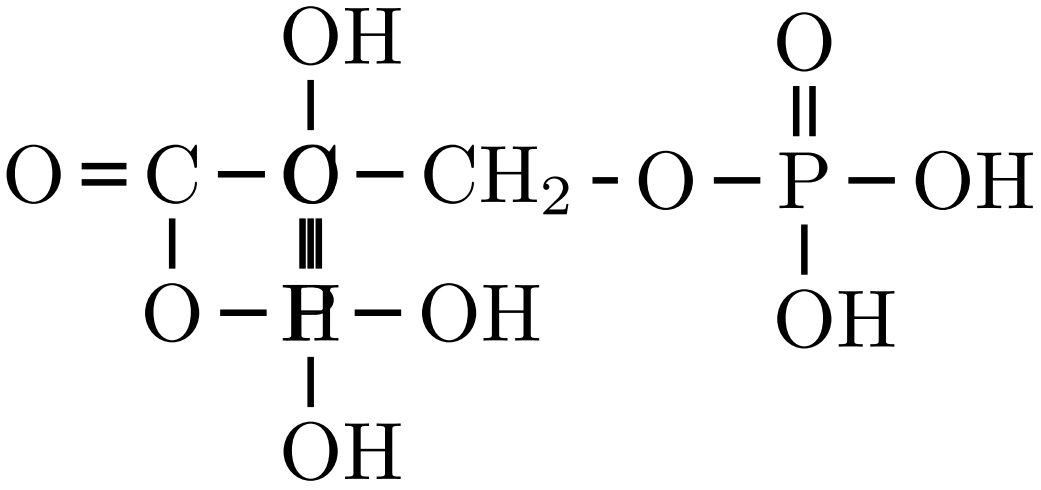
ATP produced: 0. NADH produced: 2 per glucose.

STEP 7. Phosphate transfer from 1,3-BPG to ADP

Enzyme: Phosphoglycerate kinase (PGK)

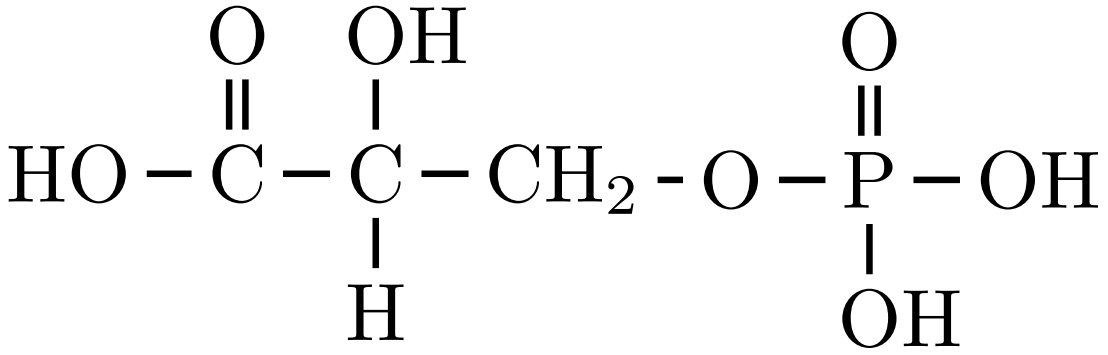
$$\text{1,3–BPG} + \text{ADP} \rightleftharpoons \text{3-PG} + \text{ATP}$$

1,3-BPG:

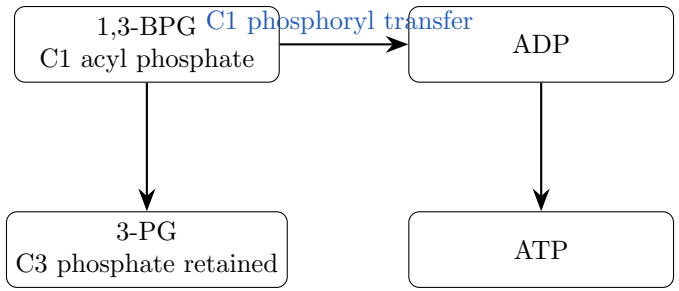


Phosphoglycerate kinase

3-phosphoglycerate:



Phosphate-transfer map



Chemical changes

1. The high-energy C1 acyl phosphate is transferred to ADP.
2. ATP is formed.
3. The C3 phosphate remains attached to the carbon skeleton.
4. The product is 3-phosphoglycerate.

Reaction classification: Substrate-level phosphorylation. There is no net oxidation-reduction reaction.

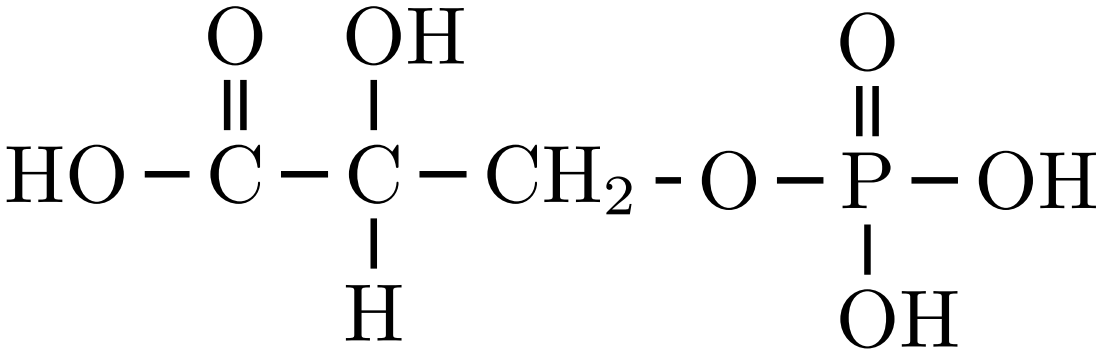
ATP produced: 2 per glucose

STEP 8. 3-phosphoglycerate → 2-phosphoglycerate

Enzyme: Phosphoglycerate mutase

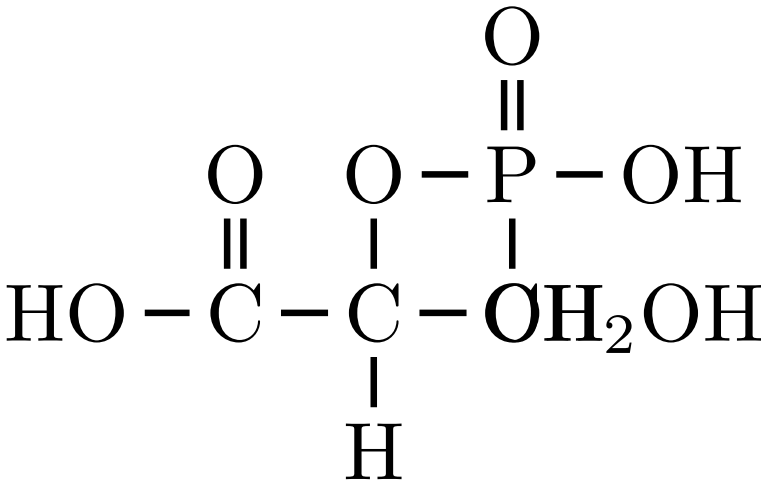
$$\text{3-PG} \rightleftharpoons \text{2-PG}$$

3-phosphoglycerate:



Phosphoglycerate mutase

2-phosphoglycerate:



Chemical changes

1. The phosphate is relocated from C3 to C2.
2. The enzyme can use a transient bisphosphorylated intermediate.
3. The final product contains the phosphate on C2.

$$\boxed{\text{C3}-\text{OPO}_3 \longrightarrow \text{C2}-\text{OPO}_3}$$

The phosphate is rearranged, not added from ATP.

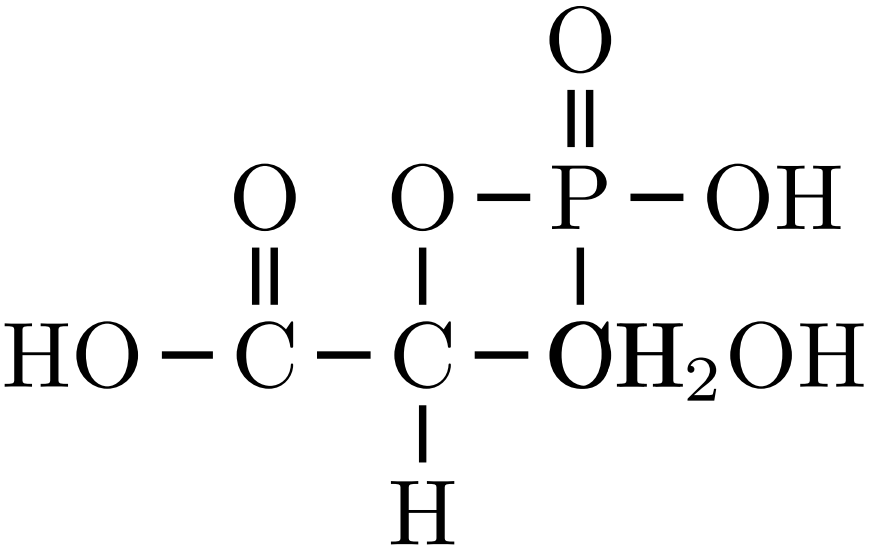
ATP consumed or produced: 0.

STEP 9. Dehydration of 2-phosphoglycerate

Enzyme: Enolase

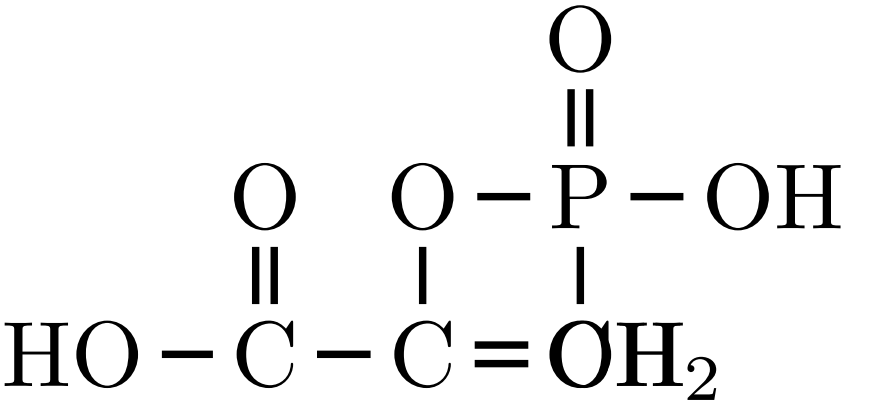
$$\text{2-PG} \rightleftharpoons \text{PEP} + \text{H}_2\text{O}$$

2-phosphoglycerate:





Phosphoenolpyruvate:



Chemical changes

1. Water is eliminated from the substrate.

2. The carbon skeleton develops a carbon-carbon double bond.
3. The phosphate remains attached to the enol oxygen.
4. PEP is produced.

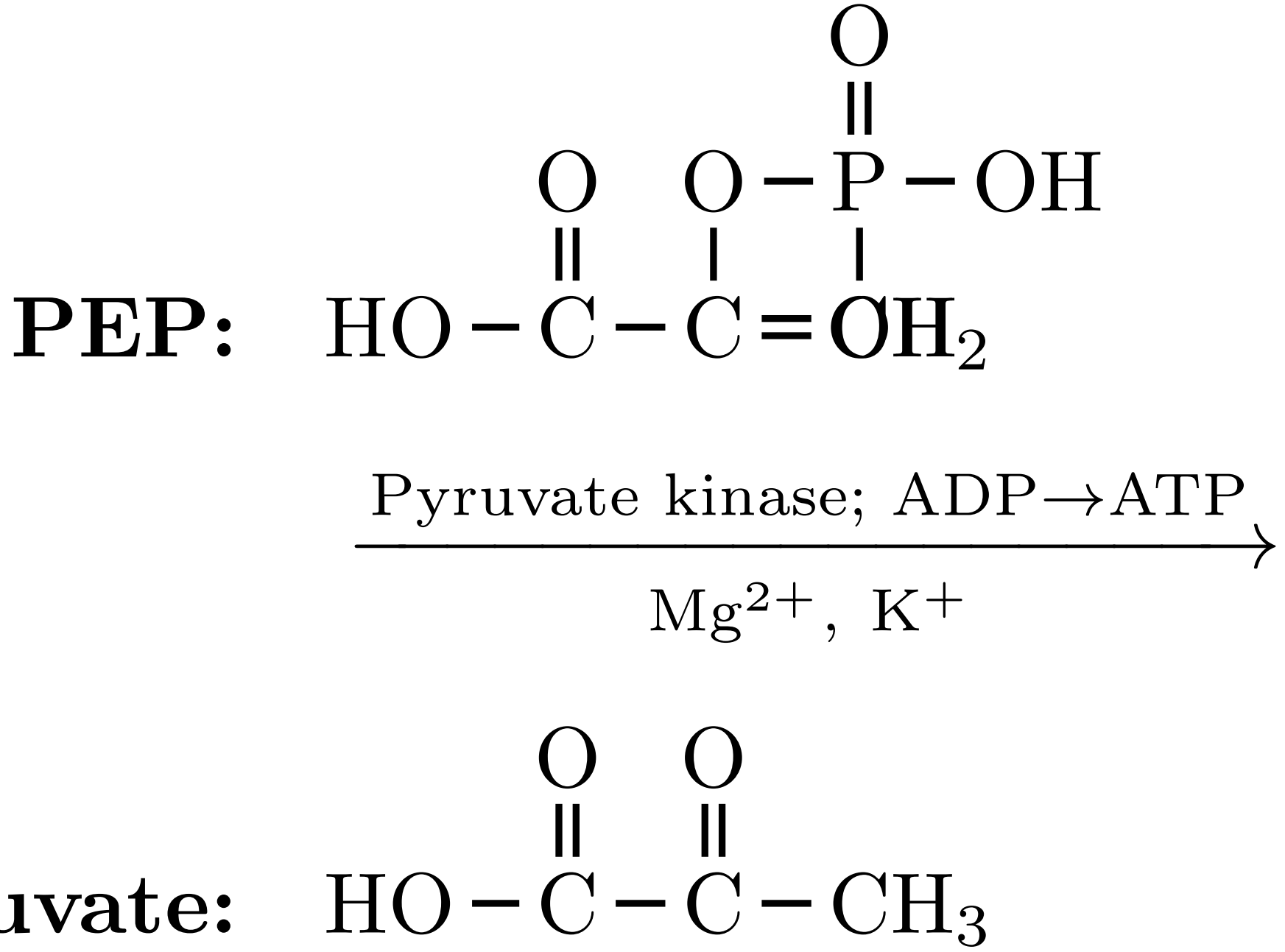
$$\boxed{\text{2-PG} \longrightarrow \text{PEP} + \text{H}_2\text{O}}$$

Electron movement: Bond electrons are redistributed during the elimination reaction to create the double bond.
ATP produced: 0.

STEP 10. Phosphate transfer from PEP to ADP

Enzyme: Pyruvate kinase

$$\text{PEP} + \text{ADP} + \text{H}^+ \longrightarrow \text{Pyruvate} + \text{ATP}$$



Chemical changes

1. The phosphate group of PEP is transferred to ADP.

2. ATP is produced.
3. The three-carbon skeleton forms the enol form of pyruvate.
4. Keto-enol tautomerization yields the more stable keto form of pyruvate.

Transfer:

$$\text{PEP} + \text{ADP} \longrightarrow \text{Pyruvate} + \text{ATP}$$

The favorable tautomerization of the pyruvate product contributes to the thermodynamic favorability of this reaction.

ATP produced: 2 per glucose

COMPLETE PATHWAY ACCOUNTING

All ten reactions

Step	Enzyme	Reaction	ATP per glucose
1	Hexokinase	Glucose → glucose-6-phosphate	−1
2	Phosphoglucose isomerase	Glucose-6-phosphate → fructose-6-phosphate	0
3	Phosphofructokinase-1	Fructose-6-phosphate → fructose-1,6-bisphosphate	−1
4	Aldolase	Fructose-1,6-bisphosphate → GAP + DHAP	0
5	Triose phosphate isomerase	DHAP → GAP	0
6	GAP dehydrogenase	GAP → 1,3-BPG; NAD^+ → NADH	0
7	Phosphoglycerate kinase	1,3-BPG → 3-PG; ADP → ATP	+2
8	Phosphoglycerate mutase	3-PG → 2-PG	0
9	Enolase	2-PG → PEP + water	0
10	Pyruvate kinase	PEP → pyruvate; ADP → ATP	+2

Net reaction

$$\begin{aligned}&\mathrm{Glucose} + 2\,\mathrm{NAD^+} + 2\,\mathrm{ADP} + 2\,\mathrm{P}_i \\ &\qquad\longrightarrow 2\,\mathrm{Pyruvate} + 2\,\mathrm{NADH} + 2\,\mathrm{H^+} + 2\,\mathrm{ATP} + 2\,\mathrm{H_2O}\end{aligned}$$

The displayed net reaction uses a common biochemical protonation convention.

Final balance

Quantity	Net result
Glucose consumed	1
Pyruvate produced	2
ATP consumed	2
ATP produced	4
Net ATP	2
NADH produced	2
Carbon dioxide released	0
FAD consumed	0

Key distinctions

1. Steps 1 and 3 consume ATP to phosphorylate sugars.
2. Step 6 oxidizes GAP and reduces NAD^+.
3. Step 6 incorporates inorganic phosphate into 1,3-BPG.
4. Step 7 transfers the C1 phosphate of 1,3-BPG to ADP.
5. Step 8 moves the phosphate from C3 to C2.
6. Step 9 removes water and forms PEP.
7. Step 10 transfers the phosphate of PEP to ADP.
8. Steps 7 and 10 are the ATP-generating reactions.